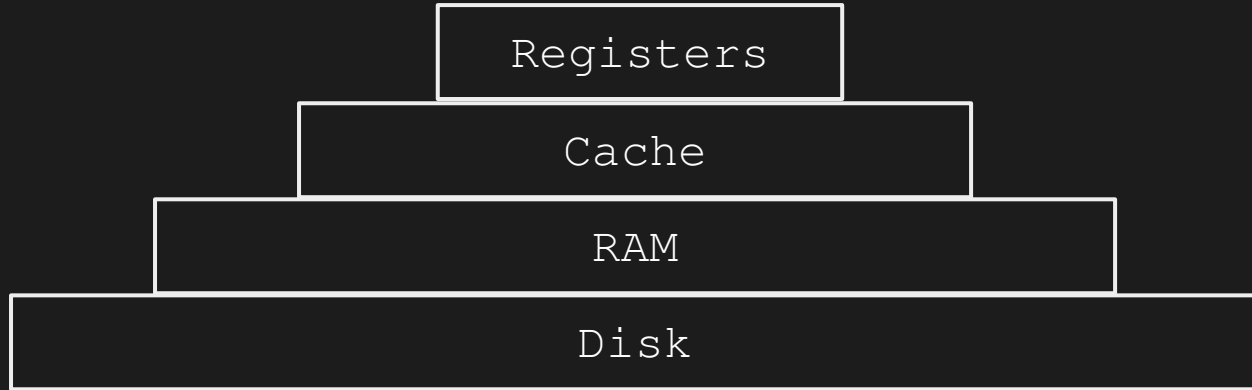


Memory Organization

The Hierarchy



What's in the Memory

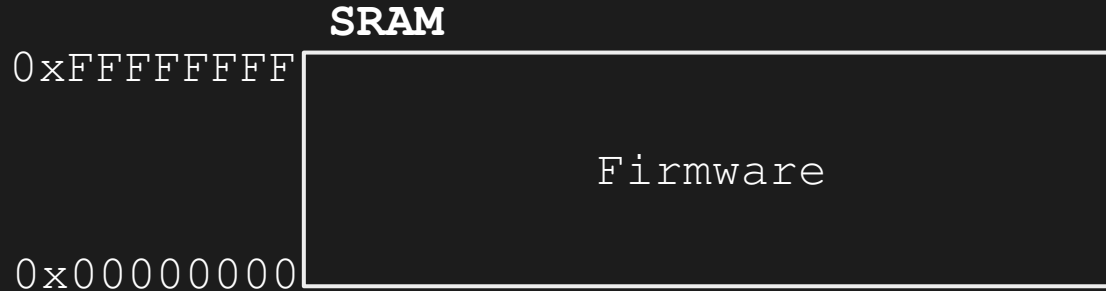
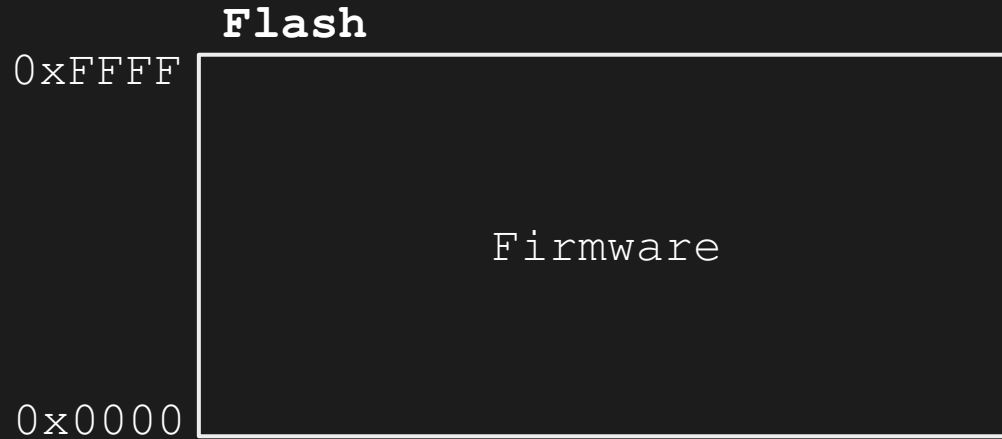
- [*] all types of variables
- [*] all functions
- [*] everything

```
int x = 4;
char *s = "abc";
short a[] = { 2, 3, 6 };
```

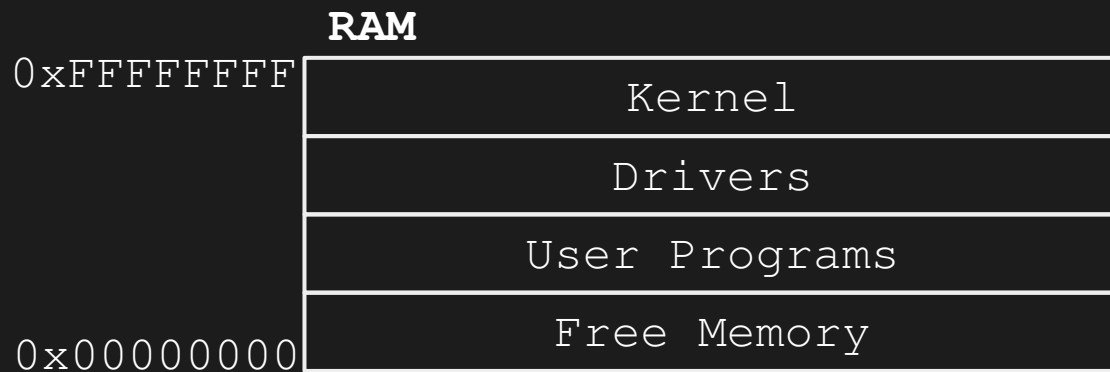
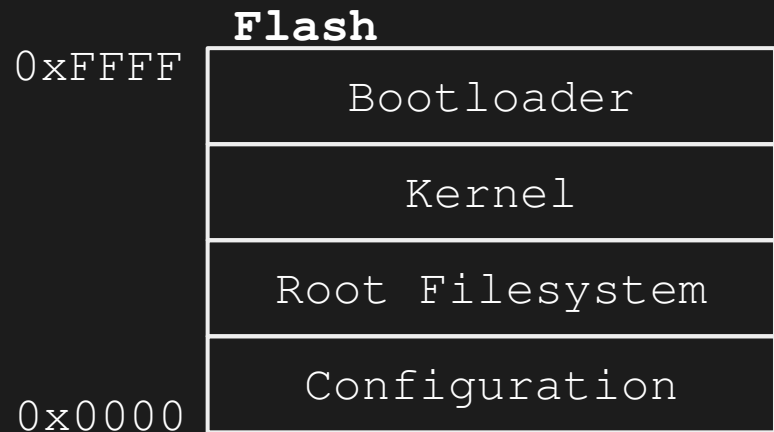
```
int
func(void)
{
    return 0;
}
```

```
// RISC-V 64bit
// func:
//      li    a0, 2      → 45 09
//      ret                    → 80 82
unsigned char code[] = {
    0x45, 0x09, 0x80, 0x82,
};
((int (*)(void)) code)();
```

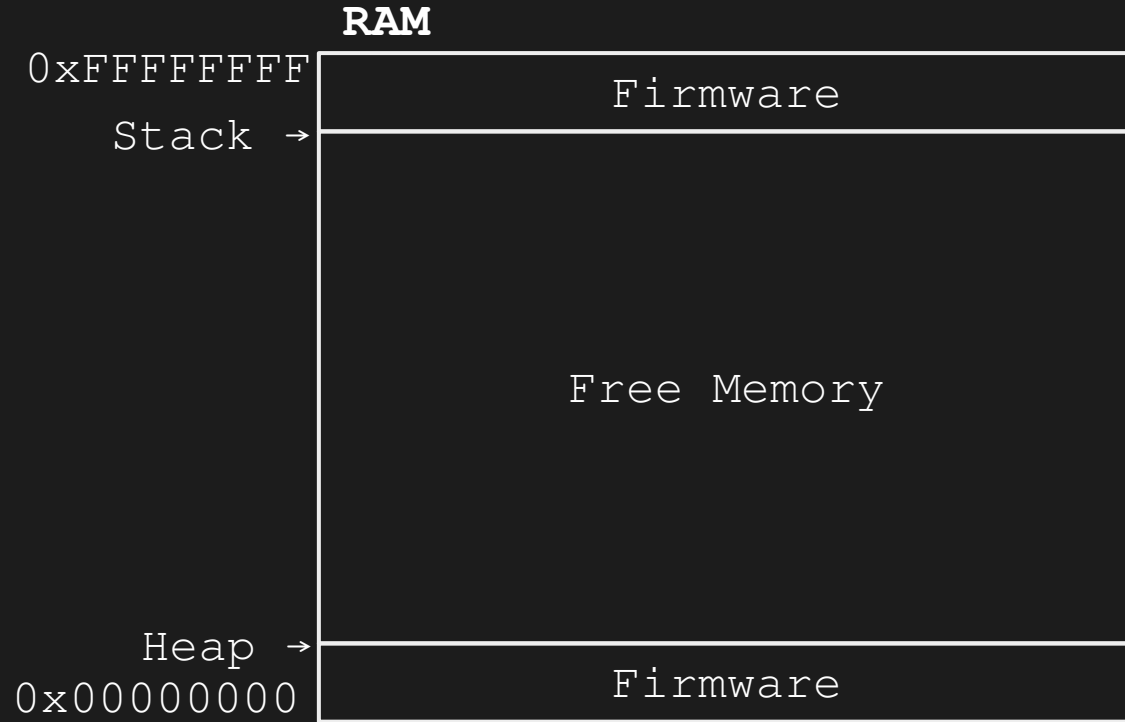
Bare Metal Layout



Embedded Layout



Free Memory



Heap vs Stack

```
// ARM64
```

```
void *mem = xalloc(...); // malloc, mmap, sbrk, ...  
//    bl      malloc  
//    str     x0, [x19]
```

```
void  
func(void)  
{  
    int x = 0;  
    int y = 1;  
}  
// func:  
//    sub     sp, sp, #16  
//    str     wzr, [sp, 12]  
//    mov     w0, 1  
//    str     w0, [sp, 8]  
//    add     sp, sp, 16  
//    ret
```

The Stack Pointer

x86

```
sub rsp, 16  
sub [rsp + 8], rax  
add rsp, 16
```

arm

```
sub sp, sp, #16  
str x0, [sp, #8]  
add sp, sp, #16
```

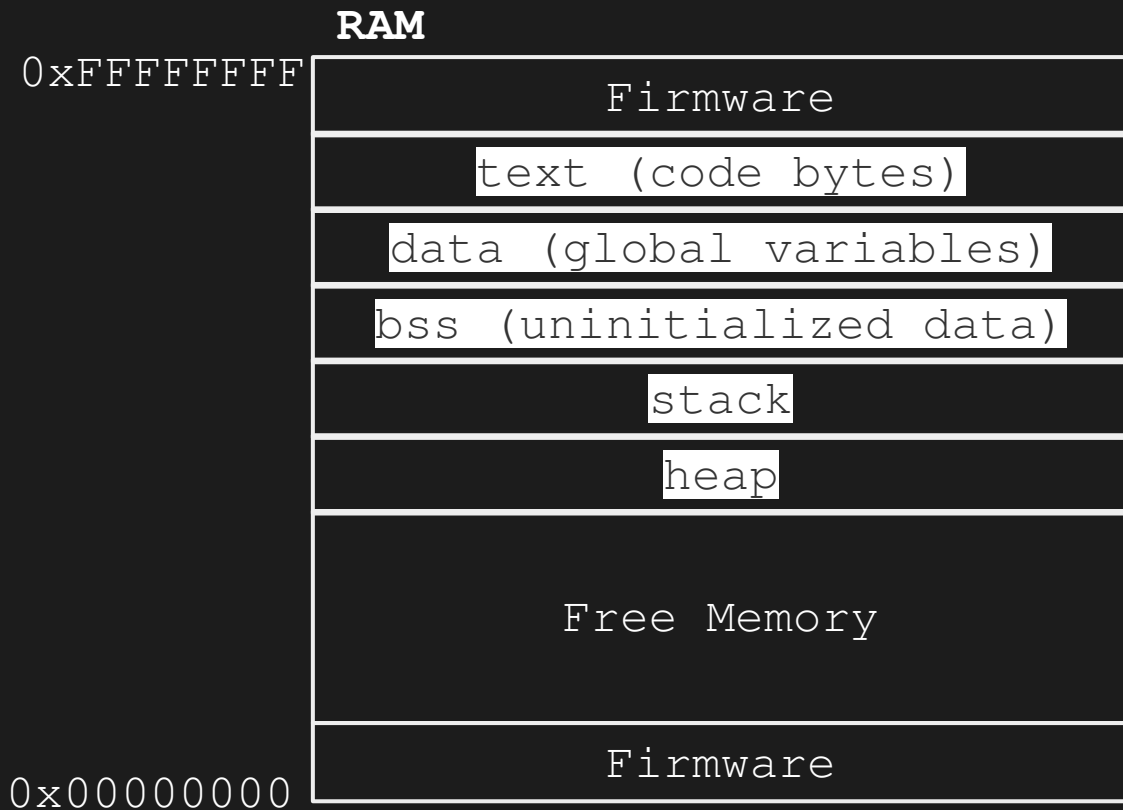
mips

```
daddiu $sp, $sp, -16  
sd      $t0, 8($sp)  
daddiu $sp, $sp, 16
```

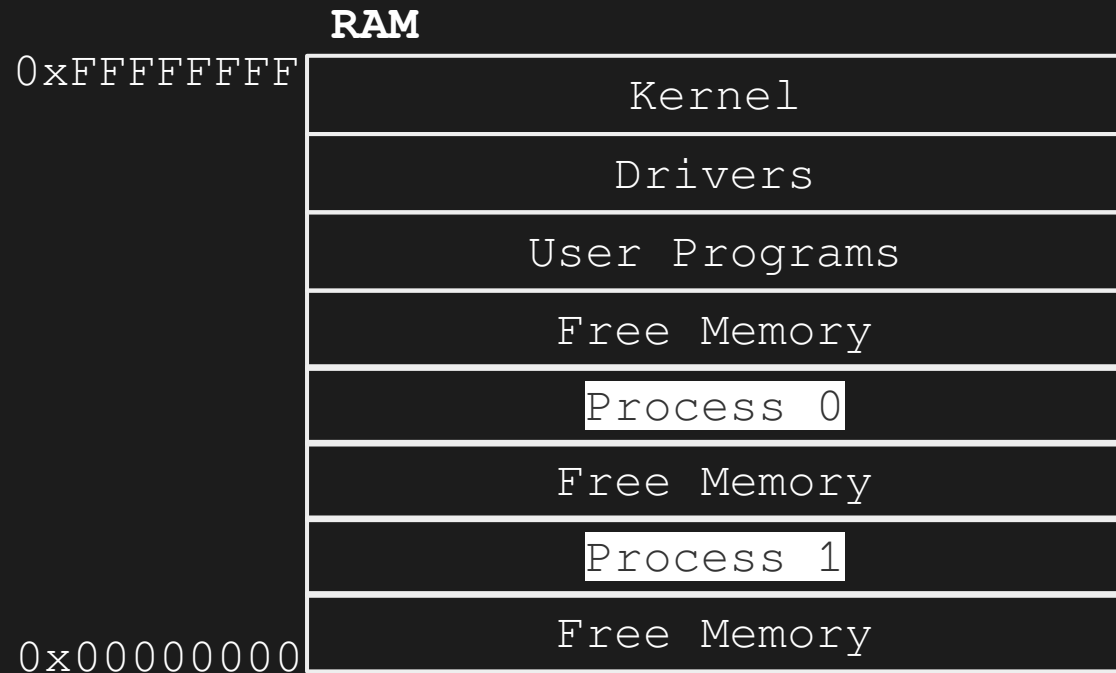
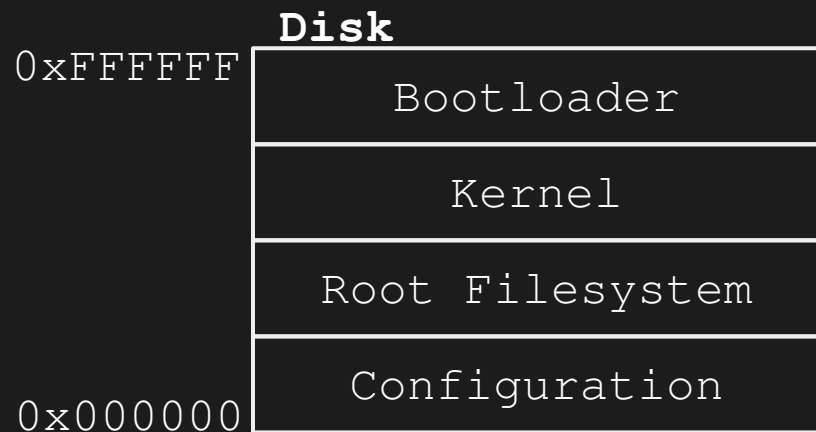
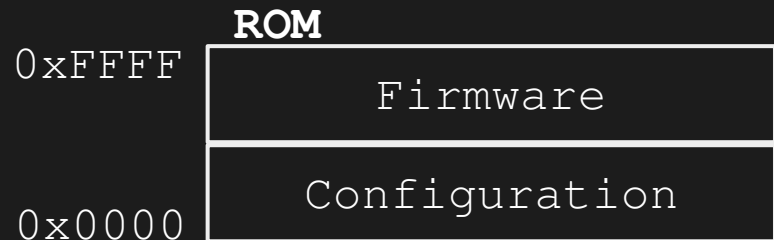
risc

```
addi sp, -16  
sd    t0, 8($sp)  
addi sp, 16
```

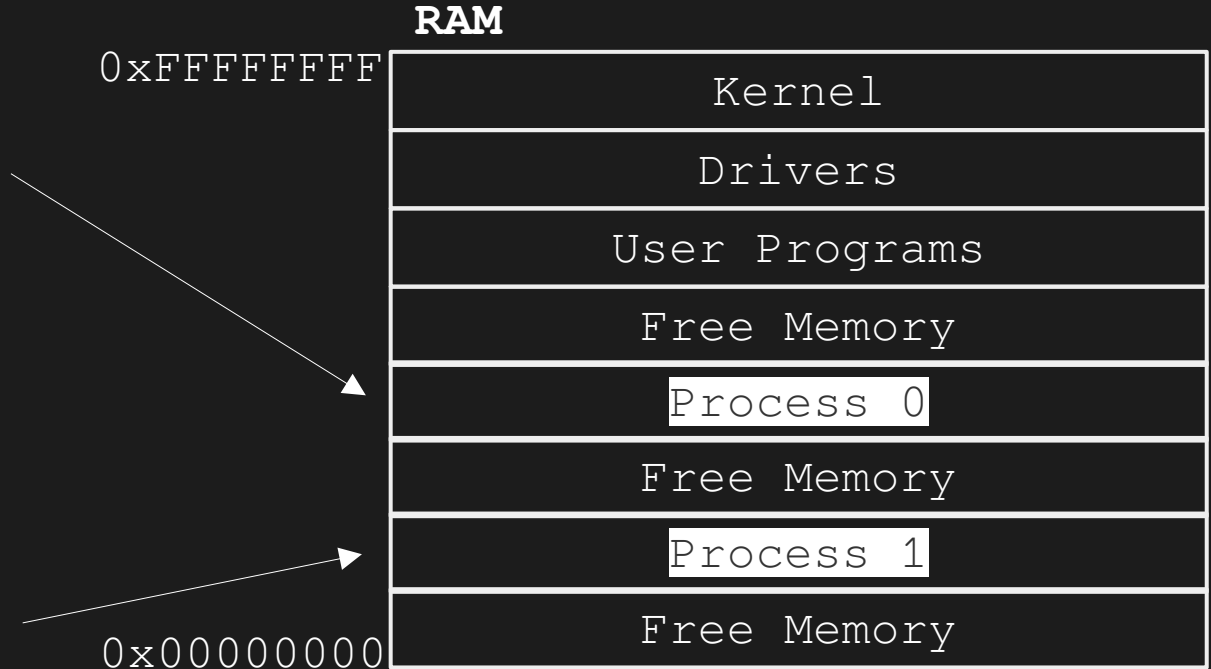
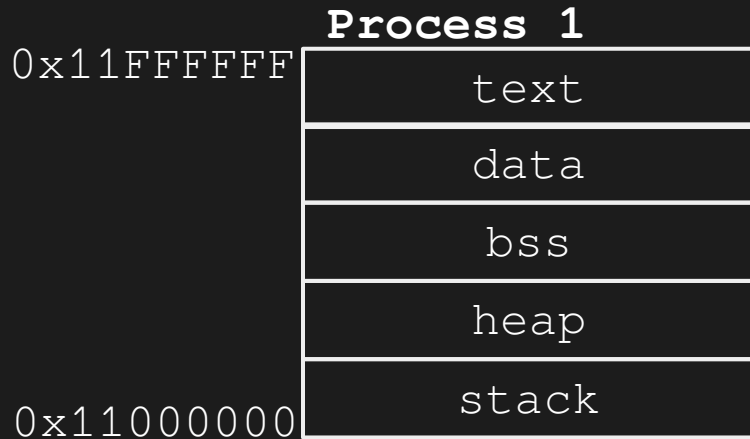
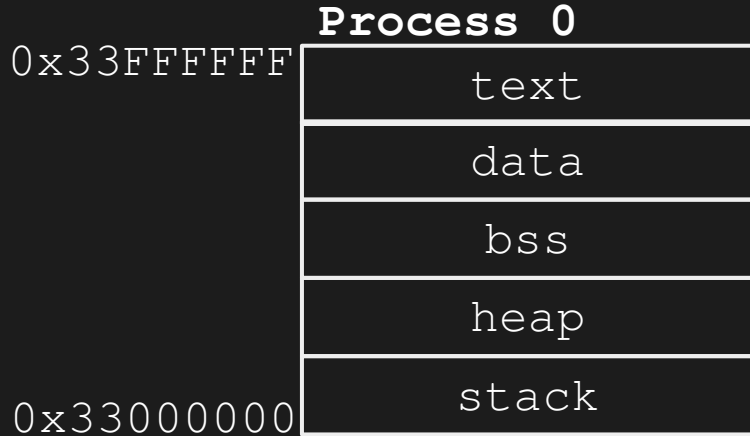
Sections



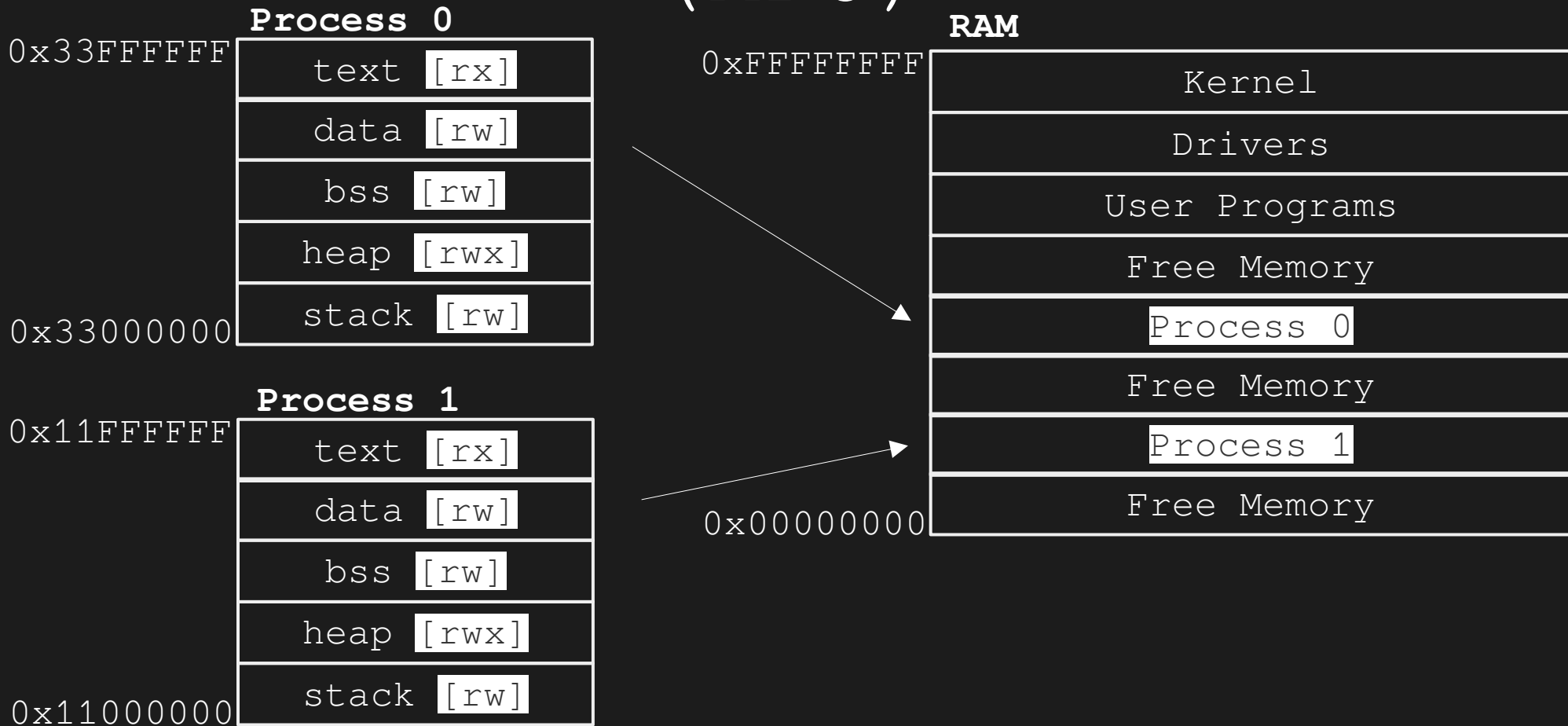
Modern Layout



Process Memory Layout



Memory Protection Unit (MPU)



Physical Memory Access

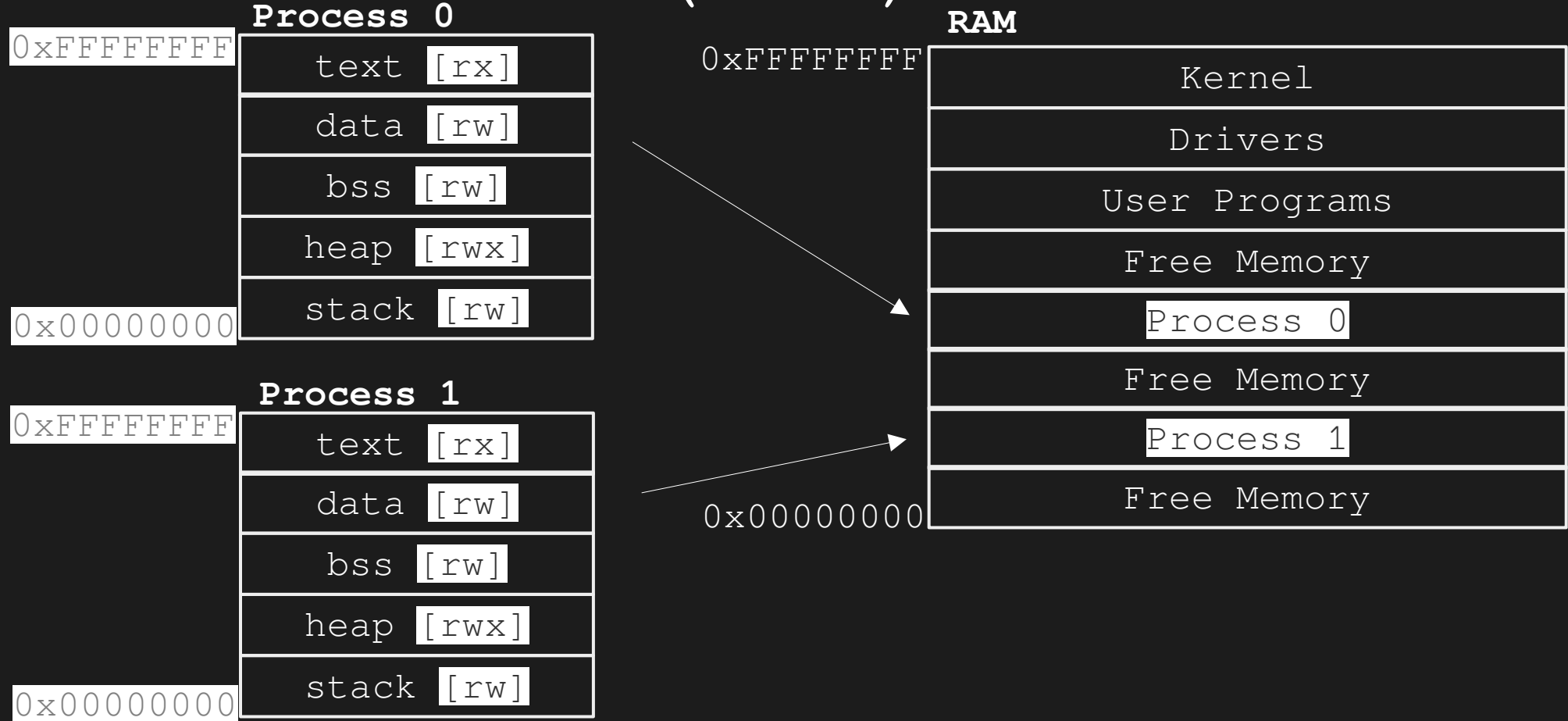
```
// Process 0
```

```
void *mem = xalloc(...); // 0x33100000  
mem = ...;
```

```
// Process 1 (Malware)
```

```
int leak = *(int*) 0x33100000;
```

Memory Management Unit (MMU)



Verifying the Theory

```
$ ps -ef # list all processes to grab a [pid]
```

```
$ sudo cat /proc/iomem # system mapping
```

```
$ cat /proc/[pid]/maps # get the memory regions
```

```
$ sudo dd if=/proc/[pid]/pagemap bs=8 skip=$(( [addr] >> 12)) count=1 | xxd # pfn
```

```
$ sudo hexdump -C -n 64 -s $([pfn] & ((1 << 55) - 1)) << 12)) /dev/mem
```

```
$ cat /boot/config-$(uname -r) | grep STRICT_DEVMEM
```

```
CONFIG_STRICT_DEVMEM=n
```

Questions?